Pure Mathematics with Mechanics GCE A Level 2019 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths With Mechanics 1, June 2019, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Mathematical tables (formulae booklets) and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

If $|x+1|<2$, then

a) \$-2 b) $x<-3$ or $x>1$
c) \$-3 d) $x<-2$ or $x>2$

2. Given that $x>0$, the value of x for which $\log_4 x+\log_4(6x+10)=1$ is

a) $\frac{2}{3}$
b) $\frac{1}{3}$
c) $\frac{2}{5}$
d) $\frac{3}{5}$

3. If $f(x)=x^3-2x-1$, $f(-2)=$

a) 11
b) -13
c) -5
d) -3

4. The tangent of the acute angle between the lines $y=4x-3$ and $y=x-5$ is

a) $\frac{2}{5}$
b) $\frac{2}{5}$
c) $\frac{3}{5}$
d) $\frac{3}{5}$

5. The period of the function $f(x)=3\cos 5x$ is

a) $\frac{2\pi}{5}$
b) $\frac{5\pi}{2}$
c) $\frac{2\pi}{3}$
d) $\frac{3\pi}{5}$

6. $\frac{2x}{(x-3)(x-5)} \equiv$

a) $\frac{3}{x-3} + \frac{5}{x-5}$
b) $-\frac{3}{x-3} + \frac{5}{x-5}$
c) $\frac{3}{x-3} - \frac{5}{x-5}$
d) $-\frac{3}{x-3} - \frac{5}{x-5}$

7. The function $f(x) = x^3 + 1$ is continuous in the interval $[-2, 4]$ and differentiable in the interval $(-2, 4)$. The value of the constant c, \$-2

a) $\sqrt{7}$
b) $\frac{2}{\sqrt{3}}$
c) $\sqrt{3}$
d) 2

8. An arithmetic progression has first term $\ln 2$ and common difference $\ln 4$. The sum of the first 4 terms of this progression is

a) $16 \ln 4$
b) $16 \ln 2$
c) $10 \ln 4$
d) $8 \ln 2$

9. If $x^2 + y^2 = 16x$, then $\frac{dy}{dx} =$

a) $\frac{8 + x}{y}$
b) $\frac{8 - x}{y}$
c) $-\frac{8 + x}{y}$
d) $-\frac{8 - x}{y}$

10. The radius of the circle with equation $x^2 + y^2 - 10x + 12y + 41 = 0$ is

a) $5\sqrt{2}$
b) $3\sqrt{5}$
c) $2\sqrt{5}$
d) $4\sqrt{5}$

11. A relationship between $(y-2)$ and $(x-1)$ is given by $(y-2)=ab^{(x-1)}$. With a suitable linear relationship between $(y-2)$ and $(x-1)$, the intercept of the straight line on the axis of $\log_{10}(y-2)$ is

a) $\log b$
b) $2\log b$
c) $2\log a$
d) $\log a$

12. $\int_0^1 e^{-x}\,dx =$

a) $1-e^{-1}$
b) $-1-e^{-1}$
c) $e+e^{-1}$
d) $-1+e^{-1}$

13. $\int e^{\ln x^2}\,dx =$

a) $\frac{1}{3}e^{x^3}+K$
b) $\frac{1}{3}x^3+K$
c) $\frac{x^3}{e^3}+K$
d) $\frac{2}{x}e^{\ln x^2}+K$

14. The range of values of k for which the quadratic equation $x^2+2kx+(k+6)=0$ has real and distinct roots is

a) -2 b) $k<-2$ or $k>3$
c) $-2\le k\le 3$
d) $k\le -2$ or $k\ge 3$

15. If $\cos\theta=\frac{3}{5}$ and θ is an acute angle, then the exact value of $\sin 2\theta$ is

a) $\frac{24}{25}$
b) $\frac{16}{25}$
c) $\frac{9}{25}$
d) $\frac{12}{25}$

16. A 3×3 matrix $\mathbf{M}$ is given as $\mathbf{M}=\begin{pmatrix} x & 1 & x-1 \\ 2 & 1 & 0 \\ 1 & x & 0 \end{pmatrix}$. If the determinant of $\mathbf{M}$ is zero, then

a) $2x^2-3x-1=0$
b) $2x^2+3x+1=0$
c) $x^2-3x+1=0$
d) $2x^2-3x+1=0$

17. The argument of the complex number z, where $z=\frac{1+i}{1-i}$, is

a) $\frac{\pi}{4}$
b) $\frac{\pi}{3}$
c) $\frac{\pi}{2}$
d) $\frac{\pi}{6}$

18. The point $(3, 10)$ is on the graph of the function $f(x)$ and $f'(3) = 114$. Given that $x = 3$ is a first approximate root of the equation $f(x) = 0$, using the Newton-Raphson procedure a value of a second approximation to the root of the equation is

a) 3.09
b) 2.92
c) 2.91
d) 3.08

19. A relation R defined on the set of real numbers is such that xRy and $yRx \Rightarrow x = y$. R is said to be

a) reflexive
b) antisymmetric
c) transitive
d) symmetric

20. A function y satisfies the differential equation $y\frac{dy}{dx} = x$. If $y = 1$ when $x = 0$, then

a) $y^2 = 1 + x^2$
b) $y^2 = 1 - x^2$
c) $y^2 = -1 + y^2$
d) $y^2 = -1 - x^2$

21. Given that ${}^nC_2 = {}^nC_3$, the value of the integer n is

a) 6
b) 5
c) 4
d) 3

22. Given that $x^2 - 4x - 1 \equiv (x + B)^2 + C$, the values of the constants B and C respectively are

a) 2 and 5
b) -2 and 5
c) 2 and -5
d) -2 and -5

23. Two functions f and g are such that $f(x) = \ln x$ and $g(x) = 9 - x^2$. The domain of the composite function $(f \circ g)$ is

a) $\{x : x \leq 3\}$
b) $\{x:0 c) $\{x:-3 d) $\{x : x < -3 \text{ or } x > 3\}$

24. $\lim_{x \to 3} \left(\frac{x - 3}{x^2 - 2x - 3} \right) =$

a) 0
b) $\frac{1}{3}$
c) ∞
d) $\frac{1}{4}$

25. $\frac{\cos 2x}{\cos x - \sin x} =$

a) $\sin x - \cos x$
b) $\cos x + \sin x$
c) $2\cos x + \sin x$
d) $\cos x - 2\sin x$

26. The coefficient of x^3 in the binomial expansion of $(2 - 3x)^5$ is

a) -1080
b) 1080
c) -360
d) 360

27. $\int_0^2 \frac{2x}{1 + 2x^2}\, dx =$

a) $\ln 9$
b) $\ln 81$
c) $\ln(\sqrt{3})$
d) $\ln 3$

28. Two vectors $\mathbf{a}$ and $\mathbf{b}$ are given as $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. The vector product $\mathbf{a} \times \mathbf{b}$ is

a) $3\mathbf{i} - \mathbf{j} - 5\mathbf{k}$
b) $3\mathbf{i} + \mathbf{j} - 5\mathbf{k}$
c) $3\mathbf{i} + \mathbf{j} + 5\mathbf{k}$
d) $3\mathbf{i} - 5\mathbf{j} + 5\mathbf{k}$

29. Statement p: $x^2 = 4$ and statement q: $x = 2$. The only statement below which is true is

Options as printed: A) $p \Rightarrow q$ B) $q \Rightarrow p$ C) $q \Leftrightarrow p$ D) \$p ot\Leftrightarrow q\$

30. There are 6 black cards and 4 red cards on a table. In how many ways can 4 cards be selected from the table if the number of colours in the selection must be equal?

a) 15
b) 24
c) 90
d) 60

31. The equations of the asymptotes of the graph of the curve $y = \frac{x - 2}{x^2 - 4x + 3}$ are

a) $x = 1,\ x = -3,\ y = 1$
b) $x = -1,\ x = 3,\ y = 0$
c) $x = 1,\ x = 3,\ y = 0$
d) $x = 1,\ x = 3,\ y = 1$

32. If $\sin(x + y) = x$, then $\dfrac{dy}{dx} =$

a) $\dfrac{1 - \cos(x + y)}{\cos(x + y)}$
b) $\dfrac{1 + \cos(x + y)}{\cos(x + y)}$
c) $\dfrac{\cos(x + y) + 2}{\cos(x + y)}$
d) $\dfrac{\cos(x + y) - 1}{\cos(x + y)}$

33. The relationship between two variables x and y is expressed by the values in the table below.

x	2	3	4	5
y	0.693	1.099	1.386	1.609

Using the trapezium rule, $\displaystyle\int_2^5 y\,dx \approx$

a) 3.636
b) 2.394
c) 4.787
d) 3.663

34. If $z = \dfrac{1 + 2i}{3 - 4i} = a + bi$, then $z =$

Options as printed: A) $z = -\dfrac{1}{5} + \dfrac{2}{5}i$ B) $z = -\dfrac{2}{5} + \dfrac{1}{5}i$ C) $z = -\dfrac{1}{5} - \dfrac{2}{5}i$ D) $z = -\dfrac{1}{5} + \dfrac{2}{5}i$

35. The position vectors of the points A and B are $\overrightarrow{OA} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$, $\overrightarrow{OB} = -2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$. A vector equation of the line passing through A and B is

a) $\mathbf{r} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k} + \mu(4\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$
b) $\mathbf{r} = -4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k} + \mu(2\mathbf{i} + \mathbf{j} - 4\mathbf{k})$
c) $\mathbf{r} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k} + \mu(-4\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$
d) $\mathbf{r} = -4\mathbf{i} + 3\mathbf{j} - 2\mathbf{k} + \mu(2\mathbf{i} + \mathbf{j} - 4\mathbf{k})$

SECTION B: MECHANICS

A particle moves so that after t seconds its position vector is $\mathbf{r}$, where $\mathbf{r} = \left[(4t - 1)\mathbf{i} + t^2\mathbf{j} + (15t - t^3)\mathbf{k}\right]$ m. The speed of the particle after 3 seconds is

a) $(4\mathbf{i} + 6\mathbf{j} - 12\mathbf{k})\ \mathrm{m\,s^{-1}}$
b) $(3\mathbf{i} + 6\mathbf{j} - 12\mathbf{k})\ \mathrm{m\,s^{-1}}$
c) $\sqrt{189}\ \mathrm{m\,s^{-1}}$
d) $14\ \mathrm{m\,s^{-1}}$

37. Three particles of weight $7\ \mathrm{N}$, $9\ \mathrm{N}$ and $4\ \mathrm{N}$ have position vectors $(4\mathbf{i}+\mathbf{j})\ \mathrm{m}$, $3\mathbf{j}\ \mathrm{m}$ and $(3\mathbf{i}+4\mathbf{j})\ \mathrm{m}$ respectively. The position vector of their centre of mass is

a) $\left(\frac{7}{3}\mathbf{i}+\frac{8}{3}\mathbf{j}\right)\ \mathrm{m}$
b) $\left(2\mathbf{i}+\frac{5}{2}\mathbf{j}\right)\ \mathrm{m}$
c) $(40\mathbf{i}+50\mathbf{j})\ \mathrm{m}$
d) $\left(\frac{7}{20}\mathbf{i}+\frac{8}{20}\mathbf{j}\right)\ \mathrm{m}$

38. Two fair dice are thrown together. The probability that the product of the scores will be at least 20 is

a) $\frac{1}{4}$
b) $\frac{2}{3}$
c) $\frac{1}{9}$
d) $\frac{2}{9}$

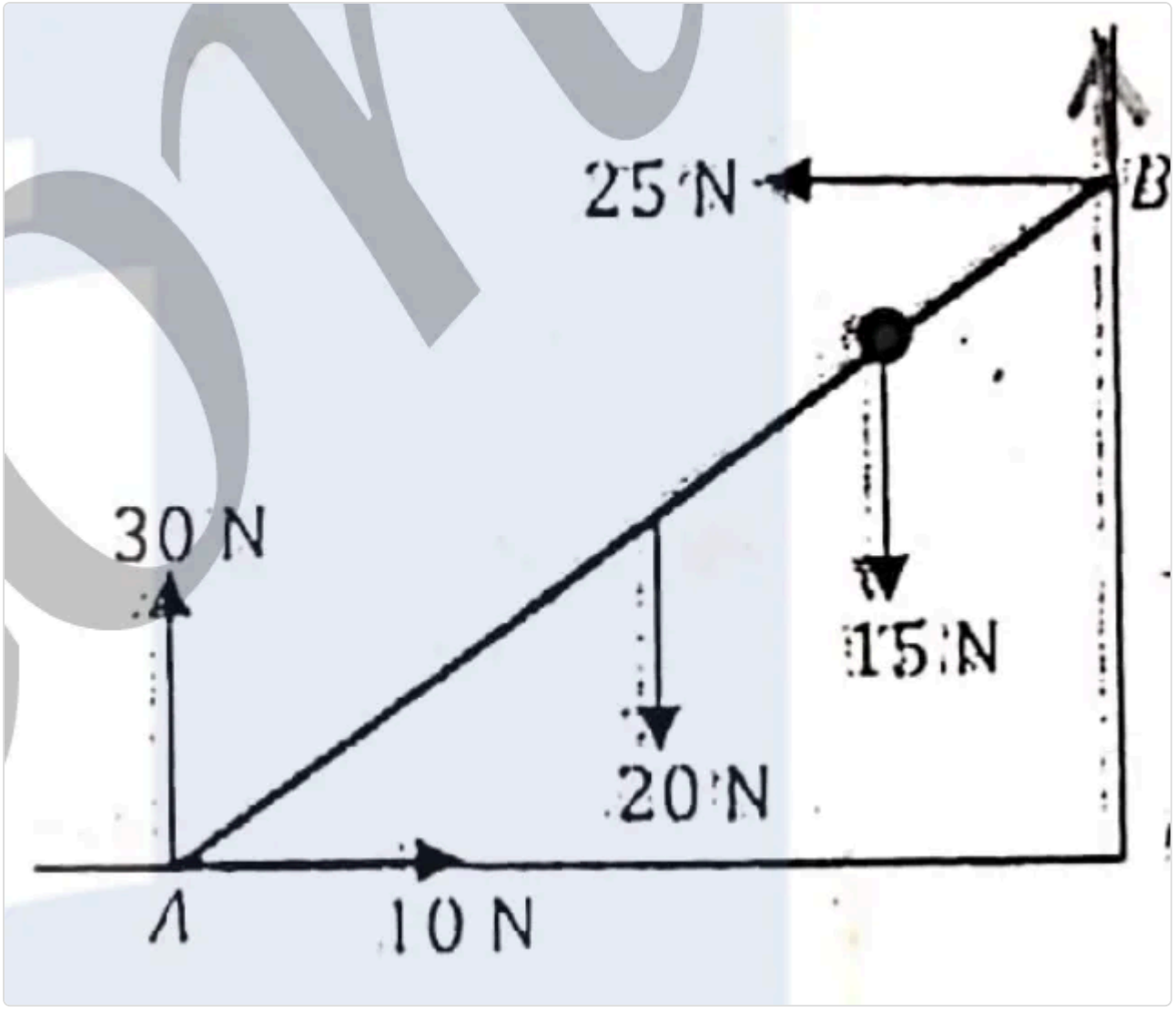
39. A man travelling due East with speed of $8\ \mathrm{km\,h^{-1}}$ watches a boy cycling along a straight road. The boy appears to be cycling due South with speed of $6\ \mathrm{km\,h^{-1}}$. The actual speed of the boy is

a) $(8\mathbf{i}-6\mathbf{j})\ \mathrm{km\,h^{-1}}$
b) $14\ \mathrm{km\,h^{-1}}$
c) $2\ \mathrm{km\,h^{-1}}$
d) $10\ \mathrm{km\,h^{-1}}$

40. A particle moving in a horizontal circle of radius $0.5\ \mathrm{m}$ has a speed $12\ \mathrm{m\,s^{-1}}$. The angular speed of the particle is

a) $24\ \mathrm{rad/min}$
b) $60\ \mathrm{rad/s}$
c) $24\ \mathrm{rad/s}$
d) $60\ \mathrm{rad/min}$

41. Fig. 1 shows a uniform ladder AB with a load on it. It is leaning against a rough vertical wall. The forces acting on the ladder are as shown.



The coefficient of friction between the ladder and the wall is

a) 0.2
b) 0.4
c) 0.2 N
d) 0.4 N

42. A car of mass 1000 kg travelling along a straight path accelerates uniformly from $15\ \mathrm{m\,s^{-1}}$ to $25\ \mathrm{m\,s^{-1}}$ in a distance of 320 m. The resistance to the motion of the car is 145 N. The driving force of the engine of the car is

a) 480 N
b) 207.5 N
c) 770 N
d) 760 N

43. A particle of mass 15 kg rests on a smooth horizontal table. It is connected by a light inextensible string passing over a smooth pulley fixed at the edge of the table to another particle of mass 10 kg which hangs freely. Taking g as $10\ \mathrm{m\,s^{-2}}$ and given that the acceleration of the system is $4\ \mathrm{m\,s^{-2}}$ when the system is released from rest, the force exerted by the string on the pulley is

a) $60\sqrt{2}$ N
b) $120\sqrt{2}$ N

c) $60\ \text{N}$
d) $120\ \text{N}$

44. The work done when the point of application of a force $\mathbf{F} = (15\mathbf{i} + 7\mathbf{k})\ \text{N}$ moves through a displacement $\mathbf{r} = (10\mathbf{i} + 3\mathbf{j})\ \text{m}$ is

Options as printed: A) $171\ \text{J}$ B) $21\ \text{J}$ C) $46\ \text{J}$ D) $21\ \text{kJ}$

45. Three particles of masses $5\ \text{kg}$, $3\ \text{kg}$ and $2\ \text{kg}$ are placed at the points A, B and C whose position vectors are $(\mathbf{i} + 3\mathbf{j})\ \text{m}$, $(7\mathbf{i} - 3\mathbf{j})\ \text{m}$ and $(-3\mathbf{i} + 2\mathbf{j})\ \text{m}$ respectively. The position vector of their centre of mass is

a) $(-\mathbf{i} + 2\mathbf{j})\ \text{m}$
b) $(2\mathbf{i} - \mathbf{j})\ \text{m}$
c) $(\mathbf{i} + 2\mathbf{j})\ \text{m}$
d) $(2\mathbf{i} + \mathbf{j})\ \text{m}$

46. Given that A and B are two independent events such that $P(A) = \dfrac{3}{10}$ and $P(A \cup B) = \dfrac{4}{5}$, $P(B) =$

a) $\dfrac{5}{7}$
b) $\dfrac{2}{3}$
c) 1
d) $\dfrac{3}{5}$

47. A particle of mass $M\ \text{kg}$ is connected by a light inextensible string passing over a fixed smooth pulley to a particle of mass $6\ \text{kg}$. Both particles are hanging freely. When the system is released and is moving in the direction of the $6\ \text{kg}$ mass with acceleration $2\ \text{m}\,\text{s}^{-2}$, and taking g as $10\ \text{m}\,\text{s}^{-2}$, the value of M is

a) $6\ \text{kg}$
b) $4\ \text{kg}$
c) $8\ \text{kg}$
d) $2\ \text{kg}$

48. Two bodies of mass m and $4m$ are moving with equal momentum. The ratio of their kinetic energies is

a) $1 : 4$
b) $4 : 1$
c) $1 : 16$
d) $1 : 2$

49. A force $\mathbf{F} = (2\mathbf{i} - 6\mathbf{j} - \mathbf{k})\ \text{N}$ displaces a particle on a plane from one point with position vector $\mathbf{r} = 4\mathbf{j}\ \text{m}$ to another point with position vector $\mathbf{r} = (3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k})\ \text{m}$. The work done by $\mathbf{F}$ is

a) $4\ \text{J}$
b) $11\ \text{J}$
c) $17\ \text{J}$
d) $23\ \text{J}$

50. A bar AB of length $0.6\ \text{m}$ and weight $3\ \text{N}$ is suspended horizontally by two strings at A and B. Given that its centre of gravity is $0.4\ \text{m}$ from A, the tension in the string at A is

a) 0.2 N
b) 1.4 N
c) 0.8 N
d) 1.0 N

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